

JAY SHAH

+1 (817) 449-3968 · shahjay15102002@gmail.com · linkedin/jay-shah · github/jayshah13 · M.S. CS · GPA 3.90 · May 2026

SUMMARY

Entry-level Python Developer with M.S. Computer Science and **3+ years of hands-on Python experience** across backend development, data engineering, automation, and AI integration. Proficient in **FastAPI, Django, Flask, Pandas, NumPy, SQLAlchemy, and PyTorch**. Experienced deploying **Python applications on AWS** with **Docker** and **CI/CD pipelines**, writing clean object-oriented code, and building solutions that are reliable, maintainable, and scalable.

WORK EXPERIENCE

University of Texas, Arlington

Aug 2024 – May 2026

Graduate Assistant — Python Development and AI Systems

Arlington, TX

- 60+ software projects had no shared Python development standard – code was inconsistent, there were no automated tests, and failures were caught manually after they had already caused problems.
- Wrote modular, object-oriented **Python** automation tooling; built **SQL** validation scripts using **SQLAlchemy**; implemented **pytest** unit and integration test suites; created monitoring dashboards with **Plotly**; documented all code and systems following clean code and **SDLC** best practices.
- 30% fewer failures, 25% faster** review cycles – well-structured, tested Python code that any developer could read, maintain, and extend without needing additional context.

Tata Consultancy Services

Jan 2024 – May 2024

Python Developer Intern — Backend and ML

Gandhinagar, India

- Production **Python** backend services processing **500K+ records** had broken integrations, silent bugs corrupting outputs, and no logging or monitoring to catch failures in real time.
- Built **FastAPI** REST services with clean OOP architecture; wrote **Python + SQLAlchemy** data transformation layers; integrated **ML models** via **PyTorch**; added structured logging, input validation, and error handling; deployed on **AWS (S3, EC2)** via **Docker**; followed **Agile Scrum** with regular code reviews.
- 2 Python services shipped** to production, system **30% more reliable**, delivery **15% faster** – clean, well-documented Python code that unblocked three teams and eliminated an entire category of runtime failures.

Celebal Technologies

May 2023 – Sept 2023

Python Developer Intern — Data and Analytics

Jaipur, India

- Three client teams needed custom **Python** solutions to automate manual workflows – **1M+ records** were processed by hand every week, scripts were one-offs with no documentation, and nothing could be reused or extended.
- Wrote reusable **Python (Pandas, NumPy)** data processing modules; built **SQL**-backed transformation pipelines; developed **Flask** web interfaces for business users; automated reporting workflows; wrote comprehensive docstrings and technical documentation for all modules.
- Manual workload reduced **25%**, data accuracy improved **18%** – clean, documented Python modules that client teams could run, modify, and extend independently without developer support.

PROJECTS

Python REST API Backend Service · Python · FastAPI · SQLAlchemy · PostgreSQL · Docker · AWS · GitHub Actions

Dec 2025 – Mar 2026

- A product team needed a scalable backend API to power their application – the existing Python scripts weren't production-ready, had no tests, and couldn't handle concurrent users without timing out.
- Designed and built a production **FastAPI** backend in **Python** with clean OOP architecture; implemented **SQLAlchemy** ORM with **PostgreSQL**; added **Redis** caching for performance; wrote **pytest** unit and integration test suites achieving **90%+ code coverage**; containerized with **Docker** and deployed on **AWS** with **GitHub Actions** CI/CD.
- API handled **300K+ requests** reliably, response time **40% faster** with Redis caching, zero-downtime deployments – fully tested, documented Python codebase that any developer could onboard to in under an hour.

Python Automation and Data Processing Platform · Python · Pandas · NumPy · Apache Airflow · SQL · AWS · Boto3 · Docker

Apr – Jul 2025

- Operations teams were manually running the same **Python** scripts every day across **12K+ records** – there was no scheduling, no error handling, no retry logic, and one bad run could silently corrupt downstream data.
- Refactored scripts into a modular **Python** automation platform: **Pandas + NumPy** for data processing, **Apache Airflow** DAGs for scheduling, **Celery** for async task processing, **Boto3** for **AWS S3** integration, **SQL** for storage – with full error handling, retry logic, structured logging, and **pytest** test coverage.
- Processing time **35% faster**, **90%** of errors caught and retried automatically – operations teams stopped babysitting scripts and the platform ran reliably without manual intervention.

AI-Powered Python Application with LLM Integration · Python · LangChain · FastAPI · PyTorch · PostgreSQL · Docker

Mar – May 2024

- 1M+ records** sat unused because users had no easy way to query or get insights from the data – every question required a developer to write a custom **Python** script, creating a bottleneck that slowed the whole team down.
- Built an end-to-end **Python** AI application: **LangChain + OpenAI APIs** for natural language understanding, **PyTorch** ML model for classification, **FastAPI** REST backend, **PostgreSQL** storage via **SQLAlchemy**, **Streamlit** frontend, and **Docker** deployment – all written in clean, well-tested Python with full documentation.
- Users could ask questions in plain English and get instant answers, developer query workload reduced **40%**, application handled **1M+ requests** reliably – Python codebase shipped end-to-end by one developer in one sprint.

EDUCATION

University of Texas, Arlington

Aug 2024 – May 2026

M.S. Computer Science · GPA: 3.90/4.00 · Distributed Systems · ML · Software Engineering · Cloud Computing

Arlington, TX

Charusat University

Oct 2020 – May 2024

B.S. Computer Engineering · GPA: 3.78/4.00 · Data Structures · Database Systems · OOP · Software Engineering

India

SKILLS

Python Core: Python 3 (advanced) · OOP · design patterns · clean code · type hints · decorators · generators · async/await · multiprocessing

Web Frameworks: FastAPI · Django · Flask · REST API design · GraphQL (familiar) · WebSockets · OAuth2 · JWT authentication

Data & ML Libraries: Pandas · NumPy · scikit-learn · PyTorch · TensorFlow (familiar) · Matplotlib · Seaborn · Plotly · SciPy

AI & Automation: LangChain · OpenAI APIs · Anthropic APIs · RAG · prompt engineering · LLM integration · Celery · task queues

Databases & ORM: PostgreSQL · MySQL · MongoDB · Redis · SQLAlchemy · Alembic · DynamoDB · query optimization

Data Processing: Apache Airflow · Apache Spark (PySpark) · Pandas ETL · Boto3 · AWS S3 · data pipelines · file processing

Testing & Quality: pytest · unittest · mock · TDD · 90%+ coverage · linting (flake8, black, mypy) · code review

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, SQS) · Docker · Kubernetes · GitHub Actions · CI/CD · Terraform (familiar)

Tools & Process: Git · Agile/Scrum · SDLC · Streamlit · Jupyter · VS Code · PyCharm · documentation · technical writing